LESSON GUIDE

Technology Plus

Unit 3: Hardware

1.3.1 – Storage Units

Lesson Overview:

In this lesson, students will compare common units of measure for data storage, including bit, byte, kilobyte (KB), megabyte (MB), gigabyte (GB), etc. and discuss how these relate to storage and file types.

Students will:

- Examine how computing devices store data in various units.
- Be able to compare practical uses between the different units of measure for data storage.
- Convert the different sizes of units of storage.

Guiding Question: How is data storage measured and how do those units relate to the types of files commonly used?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 60-70 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.3 Compare and contrast common units of measure.
  - Storage unit
    - Bit
    - Byte
    - Kilobyte (KB)
    - Megabyte (MB)
    - Gigabyte (GB)
    - Terabyte (TB)
    - Petabyte (PB)

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Lesson Background

Background Information:

All forms of data, whether it be text, images , videos, etc. are stored in structured units. These units start with bits, which represent the smallest piece of data and have a value of 0 or 1. As the complexity of the data increases, the storage units used increase in size. Bytes consist of a group of 8 bits and are commonly used to represent single characters, like the letter “B.” Expanding from there each unit increases by 2^{10}, or 1,024 , from the previous; i.e. , kilobytes (KB) consist of 1,024 bytes. Although CompTIA’s Tech+ stops at Petabyte, there are formally named storage unit sizes after that.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.3.1 – Storage Units • Lab Slides 1.3.1 – Lab From Bits to Gigabytes
Materials per Student	• Student Handout – 1.3.1 Storage Units • Student Notes 1.3.1 – Storage Units • Assessment 1.3.1 – Storage Units (Optional)

Cyber Terms:

Term	Definition
bit (b)	short for “binary digit”, represented as a 0 or 1 and comprises the most basic unit of information storage in computing
byte (B)	collection or unit of eight bits, sometimes subdivided into two halves referred to as a "nibble"
kilobyte (KB)	a storage unit consisting of 1,024 bytes, about the size of an email, small paragraph, or small icon
megabyte (MB)	a storage unit consisting of 1,024 kilobytes, the size of a photo taken with a smartphone
gigabyte (GB)	a storage unit consisting of 1,024 megabytes, about the size of a complete music album or standard-definition movie
terabyte (TB)	a storage unit consisting of 1,024 gigabytes, about the size of a modern personal device like a laptop or gaming system
petabyte (PB)	a storage unit consisting of 1,024 terabytes, reserved for estimating the size of large databases from major websites or government agencies

1.3.1 – Storage Units

Guiding Question: How is data storage measured and how do those units relate to the types of files commonly used?

Lesson Launch (3-5 minutes)

How Much Storage Do I Need?!

- Start by asking students if they have ever seen the message shown or had a similar issue where they were told they were out of storage. If students are familiar with gaming, there is often an issue of storing numerous game files.
- **Present the task:** Have students put the listed items in order based on size, smallest to largest.
- **Answer:** Text message, Photo, Short video, Mobile game, 4K movie, Social Media site.
- **Follow-up:** Ask students if they know some of the storage unit names or terms before moving to the estimated sizes.

From Tiny Bits to Petabyte Giants (20-25 minutes)

- Distribute Student Handout 1.3.1 – Storage Units and encourage them to fill in answers as you go through today's lesson.

Storage Unit Scaling in Use

- Often the storage unit used to reference a file or data is the one that makes the most sense, like the metaphor of measuring a long distance with miles and not feet.

Storage Units

- A **bit** (b) is the smallest unit of measurement in computing and consists of a value of either 0 or 1. Bits are usually only used when discussing binary or the lower levels of hardware functions and are not used as data storage units.
- A **byte** (B) consists of 8 or 2^3 bits. A byte would commonly be used to represent one single character.
- A single character would be represented by 1 byte or 8 bits, while the word "byte" would be 32 bits or 4 bytes.
- Note that after byte, each unit increases in size by 2^{10} or 1,024 units of the previous unit.
- A **kilobyte** (KB) is 1,024 bytes and is comparable to a small paragraph, email, or icon.
- A **megabyte** (MB) is 1,024 kilobytes and is roughly the size of a photo taken with a phone.
- A **gigabyte** (GB) is 1,024 megabytes and about the size of a standard-definition movie.
- **Terabyte** (TB) is approaching the limit of what a day-to-day user would be familiar with and consists of 1,024 gigabytes and would be the limits of a standard personal device.

- **Petabyte** (PB) moves from one single device to a large bank of data such as cloud services, government agencies, or large websites. It consists of 1,024 terabytes.

Lab: From Bits to Gigabytes (20-30 minutes)

- The lab will use the Ubuntu VM and command line interface to gain better details into the file and directory/folder structures in terms of storage unit sizes.
- Students will complete tasks such as modifying files to observe changes in size and the storage units used, observe the difference between allocated space and disk usage, and examine the available space on the machine.
- It is suggested that login information be distributed and virtual machines launched prior to the start of the lab.
- How can the commands ls, du, and df be used to view different aspects of storage on a Linux device?

  Students may respond with: "ls is used to show allocated storage space including some space that may not be currently in use, while du is used to show the actual disk usage, and df is used to view the system storage."
- Why is it important to understand file sizes and storage units when working with technology?

  Students may respond with: "You need to know how much space is available for new files or things needing to be installed. It may also provide insight on why a system is running slow or posting errors about space."
- What happens when a device runs out of storage? How can you manage disk space effectively?

  Students may respond with: "You will not be able to create or maybe even open files."
- How might cloud storage solutions change the way we think about file sizes and disk usage?

  Students may respond with: "It allows you to have additional storage that is not taking up physical space on the device."

Putting It All Together (5 - 10 minutes)

Discussion Questions

- Why do we use different units to measure different types of data?

  Students may respond with: "Units are used based on the related files size and purpose. It makes sense to use a larger storage unit for larger files to eliminate the need to use a larger value of a smaller unit, like using miles for long distances instead of feet?"
- How would understanding storage units assist in purchasing new devices?

  Students may respond with: "You would want a device that serves your storage needs, such as having a gaming computer that can store the game files. Perhaps you need a laptop for day-to-day work, and you want to keep it a cost-efficient purchase, then it needs less storage."
- How much digital storage do you take up considering all the devices and services you use?

  Students' answers will vary but may respond with: "Hundreds of gigabytes or several terabytes." Mention devices or services to account for such as phones, laptops, gaming systems, cloud drives, social media platforms, online shopping, streaming, etc."